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KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Rotational Dynamics

(English Medium)

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ROTATIONAL DYNAMICS

KEY CONCEPT

Types of motion :

1. **Translational :** If all the particles have same velocity at any given time than it is translational motion. Also rigid body moves parallel to itself in translational motion.
2. **Pure Rotational :** A rigid body is said to be in pure rotation if every particle of the body moves in a circle and the centres of all the circles lie on a straight line called the axis of rotation.
3. **Combined translation & rotational :** A general plane motion is a combination of translation and rotation.

Relation between linear & angular variables:

(A) Distance $S = r\theta$

(B) Acceleration $a_t = \frac{dV}{dt}$ (rate of change of speed) $= r\alpha$ (tangentially directed)

$a_r = \frac{V^2}{r} = \omega^2 r$ (directed towards centre)

Angular velocity of a point on rigid body wrt other point on rigid body
all points are rotating with angular velocity ω with respect to each other.

ENERGY OF A RIGID BODY ABOUT A FIXED AXIS

Gravitational P.E. $U = mg\,h_{cm}$ **Kinetic energy** $K. = \frac{1}{2} I\omega^2$

MOMENT OF INERTIA

Definition : Property of a rigid body by virtue of which it opposes change in its rotational velocity (angular velocity) is known as MI.
This is always written wrt a axis of rotation.
This plays same role in rotational motion as mass plays in translational motion
Difference between mass & MI is that mass is property of body & is independent of any reference from choosen but MI depends on the
(i) axis of rotation
(ii) shape of the body] mass depends only on these two things
(iii) size of the body
(iv) density of the material of the body

1. MI of a point mass : $I_{yy} = mr^2$

2. MI for point mass distribution : $Iyy' = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + ...$

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For continuous mass distribution

$I_{yy'} = \int .dm\ r^2$ where r is ⊥ distance of any element from axis of rotation.

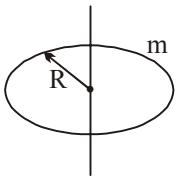
IMPORTANT POINT

MI of two or more than two bodies can be added or subtracted only when all MI are written w.r.t. same axis.

Uniform ring $I = mr^2$

(about an axis passing through centre and perpendicular to the plane.)

Disc $= \frac{MR^2}{2}$



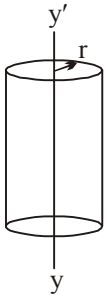
(about axis passing through centre and perpendicular to plane of disc).

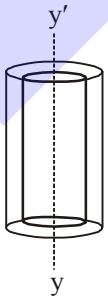
Hollow cylinder (about yy′ axis) $= mr^2$

Note : Independent of length of cylinder of same mass.

Solid cylinder (about yy′ axis)

Solid cylinder (about yy′ axis) $= \frac{MR^2}{2}$



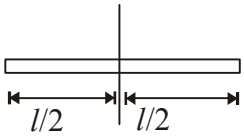


Hollow sphere (about a diameter) $= \frac{2}{3} MR^2$

Solid sphere (about a diameter) $= \frac{2M}{5} R^2$

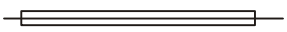
Uniform rod mass M length $\ell = \frac{ml^2}{3}$ (about axis passing through one end & perpendicular).

uniform rod $= \frac{ml^2}{12}$



(about an axis passing at $\frac{L}{4}$ from one end and perpendicular)

$I = 0$ (uniform rod (about axis passing through rod)



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Parallel-axis theorem :

Used to find moment of inertia about an axis which parallel to the axis passing through C.M.

$I_P = I_{CM} + Md^2$

$I_{CM} \rightarrow$ MI of the rigid body about an axis through CM

$I_P \rightarrow$ MI of the rigid body about an axis which is parallel to the above axis through CM & is at distance d from the axis through CM

Theorem of perpendicular axis

$I_z = I_x + I_y$

This theorem is applicable only for the laminar bodies (i.e. plane bodies). (e.g. ring, disc, not sphere)

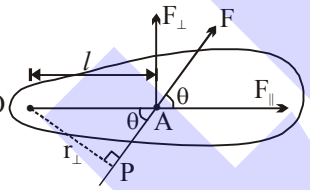
I_x & I_y are MI of body about a common pt. O in two mutually $\perp$ directions in the plane of body

I_z is MI of body about a axis $\perp$ to X & Y axis & passing through pt. O

Radius of gyration :

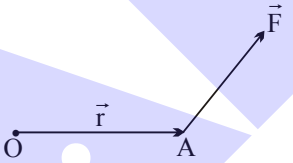
$k = \sqrt{\frac{I}{M}}$ I is MI about the axis, for which k is written.

Torque : $\tau = F\, l \sin\theta$



Vectorial representation & understanding of torque :

$\vec{\tau} = \vec{r} \times \vec{F}$



Torque due to force of gravity : $\vec{\tau} = \vec{r}_{cm} \times (M\vec{g})$

Relation between torque and angular acceleration of a rigid body :

$\tau = I\alpha$ $\vec{\tau}_{ext} = I\vec{\alpha}$ where τ and I are both written about axis of rotation.

Angular momentum of a particle : $\vec{L} = \vec{r} \times \vec{P}$

Angular momentum of a system of particle :

(a) For collection of point masses $\mathbf{L} = \Sigma \mathbf{L}_i$

(b) For rigid body $L_{axis} = I_{axis}\, \omega$

Here L_{axis} means component of angular momentum along the axis

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Relation between torque and angular momentum :

$$\vec{\tau}_{ext} = \frac{d\vec{L}}{dt} \qquad \tau_{axis} = \frac{dL_{axis}}{dt}$$

Conservation of angular momentum :

If external torque about a axis is zero then angular momentum of the system about that axis will remains conserved. So if $\tau_{axis} = 0 \Rightarrow L_{axis}$ = constant

COMBINED ROTATION AND TRANSLATION

d'Lambert's theorem says that "every possible motion of a rigid body can be represented as a combination of translational motion of the centre of mass and rotation about an axis through the centre of mass".

Kinetic Energy

$$KE = \frac{1}{2}MV_{cm}^2 + \frac{1}{2}I_{cm}\,\omega^2$$

Angular momentum

Theorem : Angular momentum of a body about any point or axis A = $\vec{L}$ of body about C.M. (parallel to axis A) + $\vec{L}$ of C.M. about axis A. $\vec{L} = \boxed{\vec{L}_{body-cm}} + \boxed{\vec{L}_{cm}}$

Instantaneous Point/Axis of rest

Axis about which combined rotation & translational motion of a body can be represented as pure rotation.

Note :

1. The axis is $\perp$ to the plane of motion 2. Can not be used for acceleration 3. $KE = \frac{1}{2}I_{IAOR}\,\omega^2$

Equilibrium & Toppling

Necessary conditions for equilibrium of an object :

1. The resultant external force must equal zero. $\Sigma\vec{F} = 0$
2. The resultant external torque about any axis must be zero : $\Sigma\vec{\tau} = 0$

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